

# AETHERIS AI

## Technical Dossier

*Architecture · Stack · AI Act-readiness · Security · Performance*

**FOR EIC ACCELERATOR · SMART&START · TECHNICAL DUE DILIGENCE**

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Aetheris AI S.r.l. · Startup Innovativa · Toscana, Italia

AI Act-ready by design (Reg. UE 2024/1689)

ISO/IEC 27001 (roadmap) · GDPR · NIS2 · Cyber Resilience Act

# 1. SYSTEM ARCHITECTURE OVERVIEW

The Aetheris AI platform is built as a layered, cloud-native architecture designed for industrial workloads. Each layer is containerised and can be deployed independently on Kubernetes (cloud or on-premise) or on edge runtimes. The architecture is AI Act-ready by design: every layer supports risk management, documentation, transparency, and human oversight out of the box.

## 1.1 High-level architecture diagram

|                |                                                    |  |
|----------------|----------------------------------------------------|--|
| PRESENTATION   | Web App · Mobile · API Gateway · SDK · IDE         |  |
| APPLICATION    | Knowledge Copilot · Vision Inspector · Predict     |  |
| ORCHESTRATION  | Agent Runtime · RAG Engine · MCP Server            |  |
| AI MODELS      | Foundation Models · Custom Models · Model Registry |  |
| DATA           | Vector DB · Feature Store · Data Lake · KG         |  |
| INFRASTRUCTURE | Kubernetes · EU Cloud · Edge · Security · Obs.     |  |

## 1.2 Design principles

- **Container-native.** All layers are containerised with Docker, orchestrated by Kubernetes.
- **Stateless-first.** Stateless APIs, stateful data layer only — enables horizontal scaling and zero-downtime deploys.
- **Audit-by-default.** All model calls, agent decisions, and data accesses are logged in a tamper-evident audit trail.
- **Multi-cloud EU.** Provider-agnostic (AWS, Azure, GCP, OVHcloud, IONOS, Arsys). Same Helm charts, different targets.
- **Edge-ready.** Critical inference paths (real-time vision, edge AI) are designed to run disconnected from the cloud.
- **Compliance-by-design.** AI Act risk classification, model registry, and lineage are first-class primitives, not afterthoughts.

# 2. AETHERIS AI PLATFORM (CORE)

The Platform layer provides the MLOps and governance substrate for all applications. It is the core of the Aetheris offering and the most defensible component over time.

## 2.1 Platform components

| Component      | Function                                         | Tech            |
|----------------|--------------------------------------------------|-----------------|
| Model Registry | Versioned catalogue of all models with metadata, | MLflow + custom |

|                       |                                                                                    |                                          |
|-----------------------|------------------------------------------------------------------------------------|------------------------------------------|
| Feature Store         | lineage, ownership<br>Reusable features with versioning and online/offline serving | Feast / custom                           |
| Vector DB             | Embeddings for RAG and semantic search                                             | Weaviate, Qdrant, pgvector               |
| Pipeline Orchestrator | DAG-based ML and data pipelines (training, retraining, evaluation)                 | Argo Workflows, Airflow                  |
| CI/CD                 | Code, model, and infrastructure as code                                            | GitHub Actions, ArgoCD, Terraform        |
| Serving               | Low-latency inference for online models                                            | FastAPI, NVIDIA Triton, TorchServe       |
| Observability         | Metrics, logs, traces; SLO-based alerting                                          | Prometheus, Grafana, OpenTelemetry, Loki |
| Data Lineage          | End-to-end traceability of data, transformations, model, decisions                 | OpenLineage, Marquez                     |

## 2.2 Governance and AI Act readiness

### AI Act risk management by design

The platform implements the AI Act risk management primitives as first-class objects: each system AI has a registered risk class, a current risk assessment, a data lineage report, a transparency documentation, and a human oversight configuration. The registry is queryable by auditors and by the AI Compliance Officer.

| AI Act article                                | Implementation                                                      |
|-----------------------------------------------|---------------------------------------------------------------------|
| Art. 9 — Risk management                      | Risk register per system AI, updated at each deployment             |
| Art. 10 — Data governance                     | Data quality checks, dataset registry, full lineage                 |
| Art. 11 — Technical documentation             | Auto-generated technical sheets per system AI                       |
| Art. 12 — Record-keeping                      | Immutable audit log of inputs, outputs, decisions, operator actions |
| Art. 13 — Transparency                        | Model cards, dataset sheets, AI Act notices for end users           |
| Art. 14 — Human oversight                     | Configurable HITL patterns: review queue, override, feedback loop   |
| Art. 15 — Accuracy, robustness, cybersecurity | Continuous evaluation, adversarial testing, security scans          |

### 3. APPLICATIONS (AETHERIS KNOWLEDGE · VISION · PREDICT)

On top of the Platform, Aetheris AI ships three focused applications in Phase 1-2. Each addresses a specific enterprise need in the Energy and Manufacturing verticals and is designed to deliver measurable ROI in 8-12 weeks.

#### 3.1 Aetheris Knowledge — GenAI / RAG / Agents

Aetheris Knowledge is the enterprise knowledge and document intelligence application. It is built on a multi-modal RAG architecture with an agent orchestration layer.

| Capability                                         | Tech                                                   | Notes                                                                  |
|----------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------|
| Document ingestion (PDF, DOCX, HTML, code, images) | Unstructured.io, custom parsers                        | OCR + layout-aware extraction                                          |
| Chunking and embedding                             | Custom chunker + BGE-M3 / E5 / OpenAI text-embedding-3 | Hybrid semantic + lexical search                                       |
| Multi-modal RAG                                    | LangChain, LlamaIndex, custom orchestrator             | Text + image + table + OCR-aware                                       |
| Agent runtime                                      | MCP (Model Context Protocol), LangGraph                | Tools, memory, planning, reflection                                    |
| Knowledge graph (optional)                         | Neo4j / Memgraph                                       | For complex enterprise domains (energy grids, manufacturing processes) |
| Safety layer                                       | NeMo Guardrails, custom validators                     | PII detection, prompt injection defense, output validation             |

#### 3.2 Aetheris Vision — Computer Vision

Aetheris Vision delivers industrial and medical computer vision with fast fine-tuning and edge deployment. It combines pre-trained foundation models with custom training pipelines.

| Capability            | Tech                                         | Use case                                             |
|-----------------------|----------------------------------------------|------------------------------------------------------|
| Object detection      | YOLOv8/v11, DINO, RT-DETR                    | Defect detection on production line, asset detection |
| Image classification  | ConvNeXt, EfficientNet, custom               | Quality grading, medical image triage                |
| Semantic segmentation | SAM 2, SegFormer, Mask2Former                | Defect localisation, medical lesion segmentation     |
| OCR + document AI     | PaddleOCR, Donut, custom                     | Technical document digitisation, form extraction     |
| Video analytics       | ByteTrack + custom event detector            | Site surveillance, process monitoring                |
| Edge deployment       | NVIDIA Jetson, Coral, ONNX Runtime, TensorRT | On-premise vision without cloud dependency           |

### 3.3 Aetheris Predict — Predictive AI

Aetheris Predict covers forecasting, anomaly detection, predictive maintenance, and energy optimisation. It combines classical statistics, traditional ML, and deep learning on time series.

| Capability              | Tech                                | Use case                                           |
|-------------------------|-------------------------------------|----------------------------------------------------|
| Time series forecasting | Prophet, N-BEATS, TFT, Lag-Llama    | Demand forecasting, energy load, production volume |
| Anomaly detection       | Isolation Forest, AutoEncoder, USAD | Process anomalies, security events                 |
| Predictive maintenance  | Survival analysis, XGBoost, LSTM    | Remaining useful life, failure prediction          |
| Energy optimisation     | Linear programming, RL              | HVAC scheduling, peak shaving                      |
| Industrial connectors   | OPC-UA, MQTT, Modbus, SCADA SDKs    | Plug into existing OT infrastructure               |

## 4. DATA ARCHITECTURE

The data layer is engineered to support three workloads: training data for AI models, operational data for AI inference, and analytical data for the customer's business. All data stays in the EU by default.

### 4.1 Data domains

| Domain                   | Storage                                   | Retention                                   | Compliance                |
|--------------------------|-------------------------------------------|---------------------------------------------|---------------------------|
| Customer documents       | Object storage (S3-compatible, EU region) | Per contract                                | GDPR, customer-controlled |
| Embeddings               | Vector DB (Weaviate, Qdrant, pgvector)    | Per contract                                | GDPR, customer-controlled |
| Structured business data | PostgreSQL, encrypted at rest             | Per contract                                | GDPR, NIS2                |
| Time series              | TimescaleDB / InfluxDB (EU region)        | Per contract                                | GDPR, customer-controlled |
| Audit logs               | Immutable log store (WORM)                | 10 years (financial), 5 years (operational) | NIS2, AI Act Art. 12      |
| Model artefacts          | Object storage + MLflow registry          | Lifetime of system                          | AI Act Art. 11            |
| Telemetry                | EU region                                 | 12 months                                   | NIS2, GDPR                |

### 4.2 Data residency and sovereignty

#### EU data sovereignty by design

All customer data is stored in EU regions by default (OVHcloud, IONOS, Arsys, or AWS eu-west / eu-central). Customers can opt for on-premise deployment for the most sensitive workloads. No data is transferred outside the EU without explicit contractual

basis and customer consent. Provider selection is documented per customer.

## 5. SECURITY AND COMPLIANCE

Security is built into every layer of the platform. The architecture is designed to satisfy the security requirements of NIS2, the Cyber Resilience Act, GDPR, and the AI Act simultaneously.

### 5.1 Security controls

| Domain                       | Controls                                                                                                                         |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Identity &amp; access</b> | OAuth 2.0/OIDC, SSO, MFA enforced, role-based access (RBAC), principle of least privilege, audit log of all access               |
| <b>Network</b>               | mTLS between services, network policies in K8s, private subnets, WAF at edge, rate limiting                                      |
| <b>Data</b>                  | Encryption at rest (AES-256) and in transit (TLS 1.3), customer-managed keys (KMS / HSM), data masking in non-prod environments  |
| <b>Application</b>           | SBOM per release, signed container images (Sigstore/Cosign), SLSA Level 3 build provenance, dependency scanning, secret scanning |
| <b>Operations</b>            | 24/7 monitoring, incident response plan, vulnerability management, regular pen testing, runbook automation                       |
| <b>AI-specific</b>           | Prompt injection defense, output validation, PII detection, model card per system, data lineage for every training run           |

### 5.2 Compliance certifications (roadmap)

| Certification                        | Target | Status    |
|--------------------------------------|--------|-----------|
| ISO/IEC 27001 (Information Security) | M18    | Roadmap   |
| ISO/IEC 42001 (AI Management System) | M24    | Roadmap   |
| SOC 2 Type II                        | M30    | Roadmap   |
| GDPR full compliance + DPIA process  | M0     | In place  |
| AI Act readiness (Articles 9-15)     | M0     | By design |
| NIS2 readiness                       | M0     | In place  |
| Cyber Resilience Act readiness       | M12    | Roadmap   |

## 5.3 Threat model and AI-specific risks

- **Prompt injection.** Adversarial inputs (crafted prompts or images to fool the model).
- **Mitigation.** Defence: input sanitization, output validation, jailbreak detectors, NeMo Guardrails.
- **Bias.** Models trained on biased data may produce discriminatory outputs.
- **Mitigation.** Defence: dataset audit, fairness metrics, human oversight, monitoring of demographic skew.
- **Data leakage.** Sensitive data may be revealed through model outputs or training data extraction.
- **Mitigation.** Defence: PII detection, output filtering, differential privacy (roadmap), data minimisation.
- **Supply chain.** A compromised component could be used as a foothold in the customer's network.
- **Mitigation.** Defence: SBOM, signed images, SLSA provenance, vendor assessment.

## 6. INFRASTRUCTURE AND DEPLOYMENT

Aetheris AI supports three deployment models: fully managed cloud, hybrid, and on-premise. The same Helm charts and IaC code target all three.

### 6.1 Deployment models

| Model                     | Use case                                                              | Components                                                       |
|---------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Cloud EU (managed)</b> | Customers who want fast time-to-value without managing infrastructure | Kubernetes on OVHcloud / IONOS / Arsys / AWS eu-region           |
| <b>Hybrid</b>             | Customers with sensitive data that must stay on-premise               | Control plane in cloud EU, data plane on customer infrastructure |
| <b>On-premise</b>         | Customers in regulated industries (healthcare, defense, energy)       | Full stack on customer-managed K8s (OpenShift, Rancher, vanilla) |

### 6.2 Edge AI

- **Hardware.** Inference runtime for Jetson (Orin, Xavier), Coral Edge TPU, and x86\_64 servers.
- **Optimisation.** ONNX Runtime, TensorRT (NVIDIA), OpenVINO (Intel) for hardware-specific acceleration.
- **Deployment.** Container-based with OTA updates; signed images; rollback capability.
- **Offline.** Disconnected operation: model + edge DB + orchestration agent run without cloud connectivity.
- **Federated.** Federated learning for cross-site training without centralising data (roadmap M18+).

## 7. EVALUATION AND QUALITY

Evaluation is a first-class component of the platform. Every model has an evaluation suite that runs continuously in production, not just at training time.

### 7.1 Evaluation infrastructure

- **Datasets.** Golden datasets per use case, maintained by the customer with Aetheris support.
- **Metrics.** Task-specific metrics (accuracy, F1, mAP, ROUGE, hallucination rate, latency).
- **Online eval.** Continuous evaluation in production (shadow mode, sampled live traffic).
- **A/B.** A/B testing framework for model and prompt iterations.
- **Regression.** Performance regression alerts; automatic rollback on degradation.

### 7.2 Target quality metrics

| Application                        | Primary metric               | Target                |
|------------------------------------|------------------------------|-----------------------|
| Aetheris Knowledge (RAG)           | Answer accuracy (human eval) | $\geq 85\%$           |
| Aetheris Knowledge (RAG)           | Hallucination rate           | $\leq 5\%$            |
| Aetheris Vision (defect detection) | Recall (defecti non persi)   | $\geq 95\%$           |
| Aetheris Vision (defect detection) | Precision (falsi positivi)   | $\geq 80\%$           |
| Aetheris Predict (maintenance)     | AUC-ROC failure prediction   | $\geq 0.85$           |
| Aetheris Predict (forecasting)     | MAPE                         | $\leq 10\%$           |
| All                                | End-to-end latency (P95)     | $\leq 2s$ per request |
| All                                | Uptime                       | $\geq 99.9\%$         |

## 8. ENGINEERING TEAM AND CAPABILITY

### 8.1 Founder technical capability

The founder is a senior AI engineer with 5+ years of production experience across the full stack of the platform. The track record is documented in the Business Plan §2 and reproduced here in summary:

| Area              | Experience                                | Production evidence                    |
|-------------------|-------------------------------------------|----------------------------------------|
| LLM and GenAI     | OpenAI, Anthropic, Mistral, open-weight   | Multi-tenant RAG systems in production |
| RAG and AI Agents | LangChain, LlamaIndex, MCP, custom agents | 10,000+ docs/day, multi-tenant         |
| Computer Vision   | YOLOv8, DINO, SAM, custom training        | +22% defect detection vs baseline      |
| MLOps             | MLflow, Kubeflow, Argo, Docker, K8s       | 99.9% uptime, 35% pipeline improvement |
| Cloud             | AWS, Azure, GCP, OVHcloud,                | 25% cloud cost reduction via           |

|         |                                          |                                  |
|---------|------------------------------------------|----------------------------------|
|         | IONOS                                    | optimisation                     |
| Edge AI | Jetson, Coral, ONNX, TensorRT            | Field-deployed vision systems    |
| Backend | Python, FastAPI, Node.js, Go, PostgreSQL | 100,000+ users/day on AI systems |

## 8.2 Open positions (Phase 1 hiring plan)

| Role                            | When           | Why critical                                            |
|---------------------------------|----------------|---------------------------------------------------------|
| Co-founder / VP Commercial      | Gate 2 (M6-M9) | B2B enterprise sales for Energy/Manufacturing verticals |
| Senior AI Engineer (product)    | Gate 2 (M6-M9) | Translates technical vision into shippable product      |
| MLOps Engineer                  | Gate 2 (M6-M9) | Reliability, observability, on-call                     |
| AI Compliance Officer (advisor) | Gate 1 (M3)    | AI Act readiness, certification roadmap                 |

## 9. TECHNICAL RISKS AND MITIGATIONS

| Risk                                               | Probability | Impact | Mitigation                                                                                              |
|----------------------------------------------------|-------------|--------|---------------------------------------------------------------------------------------------------------|
| Rapid model obsolescence                           | High        | Medium | Model-agnostic architecture, evaluation infrastructure, continuous model benchmarking                   |
| Cloud provider lock-in                             | Medium      | Medium | Multi-cloud EU support, IaC abstraction, open standards (Kubernetes, ONNX, OpenTelemetry)               |
| Dependency on third-party models (OpenAI, Mistral) | Medium      | High   | Multi-model strategy, open-weight fallback, in-house fine-tuning where economically justified           |
| AI Act risk reclassification                       | Low         | High   | Continuous regulatory monitoring, ISO/IEC 42001 preparation, modular architecture for re-classification |
| Adversarial attack on production AI                | Medium      | High   | Defense in depth, input/output validation, continuous red-teaming                                       |

|                                       |      |        |                                                                                                                                         |
|---------------------------------------|------|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Performance degradation in production | High | Medium | Online evaluation, A/B testing, automatic rollback, SLO alerting<br>Security by design, encryption, pen testing, incident response plan |
| Data breach                           | Low  | High   |                                                                                                                                         |

## 10. TECHNICAL ROADMAP (12-24 MONTHS)

The technical roadmap is structured around 5 Gates (see Business Plan §17) with explicit technical deliverables at each Gate.

| Gate   | Timing | Technical deliverables                                                                                                   |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------|
| Gate 1 | M3     | MVP Aetheris Knowledge (RAG + agent base); benchmark suite; security baseline; first dataset; container registry; CI/CD  |
| Gate 2 | M6     | MVP Aetheris Vision; SCADA/OPC-UA connectors; data lineage; model registry; SBOM; signed images; first PoC in production |
| Gate 3 | M12    | MVP Aetheris Predict; AI Act documentation automation; third-party pen test; ISO/IEC 27001 prep; HITL patterns GA        |
| Gate 4 | M18    | GA Knowledge, Vision, Predict; ISO/IEC 42001 audit; federated learning MVP; EIC application submitted                    |
| Gate 5 | M24    | Federated learning GA; CE marking for high-risk systems; SOC 2 Type II; multi-region EU; EU expansion architecture       |

## 11. APPENDIX

### 11.1 Reference architecture standards

- **CNCF.** Kubernetes API and Operator Framework for the platform layer.
- **OpenAPI.** OpenAPI 3.1 for all platform APIs; AsyncAPI for streaming.
- **ONNX.** ONNX as the canonical interchange format for models; OpenVINO/TensorRT for hardware-specific acceleration.
- **MCP.** Model Context Protocol (MCP) for agent tool-use standardisation.
- **ROS 2.** ROS 2 for future robotics integration (Phase 3).

- **OpenTelemetry.** OpenTelemetry for logs, metrics, traces; Prometheus exposition format.
- **SLSA.** SLSA Level 3 for supply chain; Sigstore for signing.

## 11.2 Glossary

- **Aetheris Platform.** Architecture organised in five layers: Presentation, Application, Orchestration, AI Models, Data, Infrastructure.
- **Applied AI.** Multi-pillar: Generative AI, Computer Vision, Robotics, Predictive AI, MLOps.
- **Aetheris Knowledge/Vision/Predict.** GenAI, CV, Predictive AI applications.
- **AI Act.** EU Regulation 2024/1689 on Artificial Intelligence.
- **RAG.** Multi-tenant reasoning system that grounds LLM outputs in a knowledge base.
- **MCP.** Model Context Protocol — open standard for AI agent tool integration.

## 11.3 References and contacts

Architecture diagrams in Mermaid / PlantUML available on request. For technical deep-dive sessions, contact: [founder@aetheris.ai](mailto:founder@aetheris.ai) · Toscana, Italia · © 2026 Aetheris AI S.r.l.